Indeterminacy, co-creation human-computer, and AI in the creative process with musical interactive systems

Abstract

This article presents a practical analysis of the application of artificial intelligence (AI) and machine learning (ML) tools in the piece "*Inconsciente Artificial*", for saxophone, bandoneon, electronics, and video (2025). The study focuses on investigating the paradigms emerging in human-machine co-creation processes within the context of artistic practice involving interactive music systems. To this end, *SOMAX2* (Fiorini and Malt 2023) software was used, enabling real-time interactions during the performance involving improvisation between human musicians and the machine. The audiovisual interactive system was developed in the *Max* environment. The experiment revealed that the degree of indeterminacy between inputs and outputs is a central paradigm in human-machine co-creation, highlighting its relevance to understanding these processes. The article outlines the theoretical framework, describes the technical and creative processes, and reflects on the outcomes in relation to human-machine co-creation concepts.

Keywords: Co-creation. Artificial Intelligence. SOMAX2. Creative process.

Introduction

This text outlines the steps and strategies employed in the creation of the piece *Inconsciente Artificial* [1], for saxophone, bandoneon, electronics and video (2025) , which incorporates artificial intelligence (AI) and machine learning (ML) tools through the *SOMAX2* [2] program. Premiered in 2025, the piece was part of *Imaginários Sonoros*[3] concert, being composed in the context of the discipline que "Composition and Performance Atelier in Music and Interactive Art" — held within "Espaço de Criação e Investigação Sonora"[4] (ECrIS) at the Centro de Pesquisa em Música Contemporânea (CPMC)[5], a complementary department of the UFMG[6] School of Music. In this environment, participants are encouraged to develop creative proposals grounded in practice-based research, integrating both artistic creation and research-oriented inquiry through a collaborative work dynamic centered on projects (Padovani 2020). Both musical direction and teaching activities were led by professors Dr. Marco Scarassatti and Dr. Rogério Vasconcelos. This work is also part of my doctoral research, which aims to investigate creative paradigms facilitated by AI technologies, with the objective of understanding and applying these resources in contexts that foster co-creative relationships between humans and machines.

The piece is motivated by a central artistic question: Could machines dream? The dream serves as a poetic allegory for musical improvisation, where the act of playing is nearly simultaneous with the emergence of ideas from the (un)conscious. The machine records the sounds performed by human musicians and then uses this sonic material to improvise. In its inability to dream, it appropriates human dreams to create its own.

In this context, and with the aim of deepening the understanding of co-creative relationships between humans and machines in artistic production, we present below a brief account of the experiences and reflections related to the piece. Part of the methodology presented is inspired by the proposal of composer and researcher Marije Baalman (2022), where some visual symbols were also used to exemplify the workflows and procedures carried out.

Theoretical Framework

Automation is increasingly shaping human life through machines, computers, and systems designed to optimize various aspects of human existence, including the arts. This shift has sparked significant discussions about the potential for machine-driven processes to replace human labor (Rouvroy and Berns 2015).

Simultaneously, a counter-narrative acknowledges the

[1] In portuguese, meaning the 'artificial unconscious' (in the psychoanalytic sense of the term).

[2] Application for musical, improvisation and composition using AI with machine listening, cognitive memory activation model, multi-agent architecture, full application interface to agent patching and control, and full Max library API. Developed by Ircam Paris. Available in: https://forum.ircam.fr/projects/detail/somax-2/

[3] Interactive music and art group, comprised of students, researchers, and professors from the Federal University of Minas Gerais, founded by José Henrique Padovani and co-directed by him, Rogério Vasconcelos, and Marco Scarassatti.

[4] Creative and Sound Investigation Space

[5] Center for Research in Contemporary Music

[6] Universidade Federal de Minas Gerais - UFMG (Federal University of Minas Gerais)

constraints and problems associated with these emerging technologies, highlighting their potential to enhance new creative endeavors (Hui 2023). Given the profound social and political implications of these developments, it is essential to critically examine and potentially redefine the role of machines in the creative process and their interactions with humans.

According to Santaella (2023), "AI challenges the notion that humans have of themselves, forcing us to seek new and more adequate concepts about ourselves" and emphasizes the need "to develop a discourse that contributes to human adaptation to the new conditions of their existence". Therefore, it is essential to understand how artists can position and adapt themselves to this reality, marked by the growing availability of tools that can create and interact with humans in the artistic process. Such a reevaluation seeks to empower artists to effectively integrate these technologies into the evolving creative landscape.

The development of what we now refer to as AI and ML has introduced new paradigms into creative interactions between humans and machines. These paradigms may serve dual purposes: improving user experiences and enhancing human capabilities across various domains. In this context, studies on "human-computer co-creativity" — (Louie et al. 2020), (Pasquier et al. 2016) and (Davis 2021) — seek to understand what it means for a machine to interact actively and proactively, even under human oversight. Such interactions often produce unpredictable outputs (Moore 2021) that extend beyond ordinary support tasks within creative processes, requiring a form of "interaction where both entities mutually influence each other within a collaborative and improvisational environment" (Davis 2021).

Davis (2021) defines a co-creative situation involving humans and machines as one in which both entities mutually influence each other within a collaborative and improvisational environment, rather than merely dividing labor. Similarly, Jia Chi(Chi 2024) highlights that AI has transformed the creative landscape by moving beyond autonomous art generation to enhancing human artistic efforts through collaborative processes.

Moore (2021) distinguishes two roles for artificial intelligence in artistic practice: AI as an assistant and AI as a collaborator. While AI assistants perform predefined and predictable tasks, allowing humans to focus on other creative parameters ("Humans in the Loop") (Wang and Bidadanure 2019), collaborative AI systems are characterized by their ability to surprise human users during the creative process. These surprises often take the form of sounds, gestures, forms, or other creative suggestions that might not have been apparent to the human artist. Moore notes:

> Pursuing these meta-goals has clarified two broad frameworks for how I think about implementing artificial intelligence in my work: 1 - AI as assistant and 2 - AI as collaborator. Assistants carry out tasks in prescribed, predictable ways, enabling humans in the performance or process to focus on other creative parameters. [...] AI collaborators, which are given more attention in this paper, are more complex systems, resisting predictability, and are capable of 'surprising' human users during the creative process with sounds, forms, gestures, connections, or other creative ideas not readily apparent to the human. These surprises or 'creative suggestions' offered by artificially intelligent collaborators can then be responded to, curated, further explored, or denied, depending on the artistic moment or goal. (Moore 2021)

For such surprises to occur, Moore argues that the system must exhibit significant complexity and limited user control. Bown et al. (2020) further suggest that active system influence within the creative process requires two conditions: first, the system must stimulate the human to think differently and embrace the novelty proposed by the machine; second, the system must make contributions that are context-sensitive and responsive to the user's iterative input. In other words, a human-machine co-creation system necessitates mutual influence between the two agents during the creative process. This underscores the growing importance of understanding how machines can exhibit "creativity" by generating indeterminate responses for humans.

In light of this brief theoretical discussion, we will now present the technical and practical aspects developed in the piece, as well as reflect on how our experience aligns with the content presented in this theoretical framework.

Improvising with SOMAX2

Main Concept

The *SOMAX2* program utilizes a generative AI model to deliver real-time machine improvisations that align seamlessly with both the internal stylistic elements of the chosen corpus and the evolving external musical environment. *SOMAX2* is capable of processing MIDI and audio inputs, managing corpus memory, and generating outputs. The model is designed for ease of use, requiring minimal setup to enable its agents to interact autonomously with musicians and among themselves. The system provides extensive manual controls for tailoring the generative process and interaction strategies, making it highly adaptable and intelligent.

The program features two main components: `Influencer Agents` and `Players` (sound corpora). During interaction, the audio recorded in the buffer is analyzed by various audio descriptors and segmented based on the identified characteristics. When a relationship is found between the input from an `Influencer` and the stored audio, the corresponding segment is triggered. This response from the `Player` can operate in either a reactive or continuous mode. The corpus can be preloaded or recorded in real-time.

Although the program comes with a ready-to-use application, we chose to utilize the available objects in a new patch created in Max, for increased flexibility and control. In *Inconsciente Artificial*, the sound corpus was generated in real time during the performance. However, the `Influencers` interacted in a crossed manner: `Influencer 1` (saxophone) interacted with the audio buffer of `Player 2` (bandoneon), and vice versa [Fig 1]. The idea was to introduce a higher degree of unpredictability in the machine's responses through the use of a sound input that diverged from the

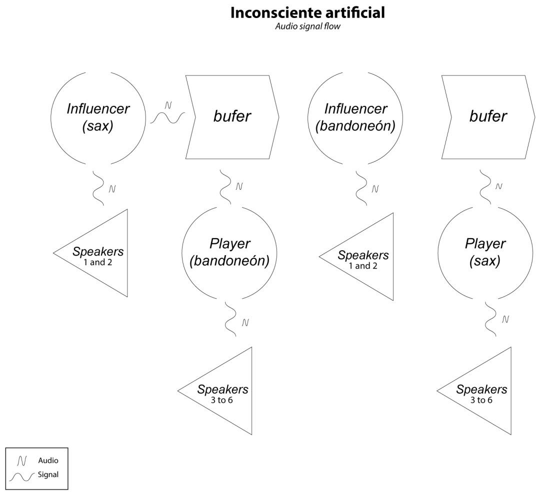

Figure 1: Audio signal flow

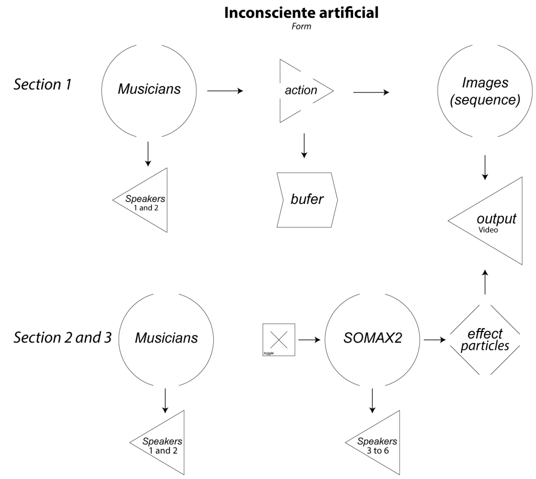

Figure 2: Form

recorded corpus. The pursuit of a certain degree of indeterminacy and complexity, as Moore argues, aimed to understand how a system can interact with and surprise human musicians during a performance. This interaction fosters a situation of co-creation, in which humans and machines collaborate creatively.

Form

The piece is divided into three distinct moments. Since the buffer recordings occurred in real time, we intended to generate diverse material capable of enabling different interactions. While the piece evokes a free improvisation, it is structured with specific sound aesthetics for each session.

In the first section [Fig 2], the focus was on exploring percussive sounds from the two instruments. The corpus recording was only activated after the first minute of interaction, and, in this initial stage, the algorithms responsible for interaction (`Players`) were not yet operational.

In the second section [Fig 2], we sought a sonic texture that blended the timbres of the saxophone and the bandoneon, using clusters and long notes. At this stage, the players were activated, but the human musicians did not react to the machine's behavior, continuing to explore a sonic atmosphere distinct from the first section. The `buffer`, with a duration of four minutes, was completely filled, marking the end of this part.

In the final section of the performance, the improvisation was guided by the machine. The piece concludes with the *SOMAX2* players interacting and performing alone, reacting to the sound captured by the room's microphones.

Video

For the production of the interactive video [Fig 4], we used *Jitter*, a module of the *Max* program designed for image manipulation. In the visual concept of the piece, our intention was to rapidly generate images representing distorted real forms, with no apparent connection between them. To achieve this, we used *ComfyUI* [7] to create at least six images that would be manipulated and integrated with the musical component.

In the first section, where there is no interaction from the *SOMAX2* players, the images were triggered based on a predefined sound intensity level (threshold) related to the amplitude of the saxophone and bandoneon. When the sound intensity exceeded this limit, an image was displayed abruptly on the screen.

In the second section, as soon as the players were activated, a static image of a sleeping woman was displayed on the screen. With each interaction from the players, this main image fragmented into particles, creating a dynamic visual effect. The images generated in response to the sound of the saxophone and bandoneon merged with the static image, progressively disintegrating based on the interaction with *SOMAX2*.

To create the particle disintegration effect, we mapped and normalized the amplitude of each player on a scale from -1 to 1, using these values as coordinates to simulate the movement of a cursor on the screen. In this way, the position of the particles on the image was directly related to the machine's interaction.

The poetic intention of the piece is mirrored in this interaction: the machine, limited by its inability to dream, transmutes human dreams to generate its own teluric expression.

Indeterminacy and Co-creation

During creation and experimentation, we explored various levels of indeterminacy that could generate sonic material to stimulate and surprise human performers during interaction with the machine. Our main challenge lay in the fact

[7] Open source, node-based program that allows users to generate images from a series of text prompts and use Stable Diffusion as base model. https://www.comfy.org

that the sound samples used by the machine were recorded from improvisations by the musicians themselves, which introduced a certain arbitrariness into the process. In other words, the recorded material was predetermined and familiar, limiting the desired unpredictability. Faced with this, we chose to stimulate the machine to react to the performers' sounds in a less obvious way, avoiding associating all the similarities identified by the descriptors. The goal was to make the sonic correspondences less clear, thereby creating a more surprising and less predictable dialogue between humans and the machine. In this sense, we believe that, based on the previously presented definitions of the concept of human-machine co-creation, indeterminacy is one of the paradigms that help us establish relationships of creative interactions involving humans and computers. The more unpredictable the sonic results we heard, the more stimulated we were to change direction or react differently from what we had anticipated. However, discussions about co-creation are not limited to understanding how the machine can surprise humans. They also involve reflecting on when we can consider a machine to be genuinely creative in its intervention. We further believe that, with the advancement of artificial intelligence technologies, the relationship with machines becomes more compelling when it transcends mere instrumentality and evolves into a collaborative and proactive partnership in the creative process. Rethinking how these "new" relationships are constructed is part of a broader discussion, of which this article presents only a fragment, related to our practical experience during the composition of the piece.

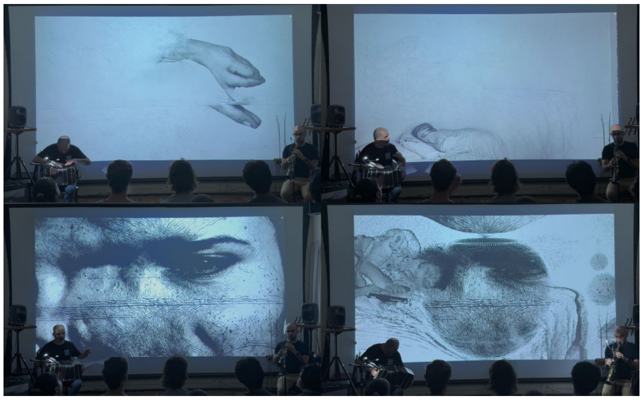

Figure 3: Video Interaction

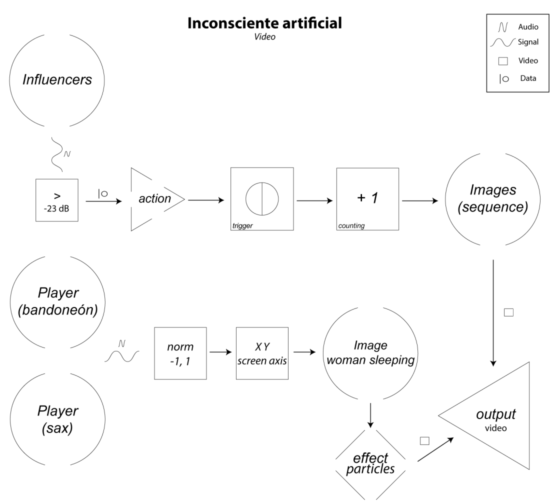

Figure 4: Video

Final Remarks

In this paper we covered the creative processes of using tools based on artificial intelligence (SOMAX2) in *Inconsciente Artificial*, and a theoretical-practical discussion on human-machine co-creation in the context of artistic production.

The main goal was to explore the paradigm of indeterminacy in co-creation processes, investigating the extent to which the machine can be considered collaborative and creative in the context of art. The arguments of Moore (2021) and Davis (2021)are corroborated, as they emphasize that more unexpected results capable of surprising humans, potentially characterizing creative behavior on the part of the machine.

As part of a broader research project, we expect the practical and reflective approach presented to contribute to discussions on the possibilities of interaction between humans and machines, especially amid advancements in artificial intelligence in the creative field.

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